$$\mathrm{v} = u \ln\left(\frac{m_0}{m}\right). \qquad (2)$$

2.

$$v_1 = \sqrt{gR}\,, \qquad (3)$$

(2)

$$m_0 = m \exp\left(\frac{v}{u}\right) = 4.87 \times 10^3\ kg. \qquad (4)$$

3. F ,

, (1)

$$mdv - dmu = Fdt\,, \qquad (5)$$

$\mu = -dm/dt$

$$m\frac{dv}{dt} = F - \mu u\,. \qquad (6)$$

,

.

$F = mg$

$$m\frac{dv}{dt} = mg - \mu u\,. \qquad (7)$$

4. $v = 0.$

$v = 0$ (7)

$$-\mu g = \frac{d\mu}{dt} u\,. \qquad (8)$$

$\mu(0) = m_0 g / u$

$$\mu(t) = \frac{m_0 g}{u} \exp\left(-\frac{gt}{u}\right)\,. \qquad (9)$$

5. $\mathrm{v}(t) = A_1 t + A_2 \ln(1 + A_3 t)$ $m = m_0 - \mu t$ (7)

$$A_1 = -g\,, \qquad (10)$$

$$A_2 = -u\,, \qquad (11)$$

$$A_3 = -\frac{\mu}{m_0}\,. \qquad (12)$$

6. .

.

$$\mu_{opt} = \infty\,. \qquad (13)$$

,

(2)

$$\mathrm{v} = u \ln\left(\frac{m_0}{m}\right). \qquad (14)$$

$$H_{\max} = \frac{u^2}{2g} \ln^2\left(\frac{m_0}{m}\right). \qquad (15)$$

7. v' .

v . w